
GeoCo-SAVi: Geometry-Consistent Slot Attention for Explicitly Editable Object Representations

Haoxiang Huang^{1,*}, Zhekai Wang^{2,*}, Xiang Liu³, Sen Cui^{3,†}, Changshui Zhang^{3,†}

¹ University of Science and Technology of China, ² Beijing Institute of Technology, ³ Tsinghua University

* Equal Contribution, † Corresponding authors

Abstract

Object-centric video models represent scenes with slots, yet exposed geometry can vary in meaning with appearance. In Invariant Slot Attention (ISA), explicit position and scale can disagree with the decoded center and extent; edits can yield unexpected motion or resizing, and replacing appearance can shift geometry.

GeoCo-SAVi promotes **geometric authority** and **semantic alignment**. Its spatially equivariant, object-wise decoder makes position and scale effective commands: changing them moves or resizes the rendered support. Factual position alignment ties position to the decoded center, and normalized attention overlap discourages duplicate allocation. Appearance transplantation aligns geometry semantics across objects, so recipient geometry governs layout while donor appearance supplies shape. A temporal initializer propagates calibrated slots across frames.

On Obj3D, GeoCo-SAVi matches ISA reconstruction, reduces p -centroid error by over 80%, and cuts appearance-induced size variation by over 50% while producing the expected translation and scale responses. On 250 MOVi-C videos, it also improves reconstruction, instance grouping, and fixed-identity geometry control over two same-protocol references. GeoCo-SAVi transforms explicit geometry into compositional control, making both position and scale more readable and editable.

Date: September 6, 2026

Correspondence: Sen Cui and Changshui Zhang at {cuis,zcs}@mail.tsinghua.edu.cn

Project: <https://GeoCo-SAVi.github.io>

Code: <https://github.com/starx237/GeoCo>

Contents

1	Introduction	3
2	Related Work	4
3	Methodology	4
3.1	Preliminary: Invariant Slot Attention	4
3.2	Spatially Equivariant Decoder	5
3.3	Phase 1: Factual Geometry Calibration	6

3.4	Phase 2: Counterfactual Appearance Transplantation	7
3.5	Training Objective and Curriculum	7
3.6	Implementation Details	7
4	Experiments	9
4.1	Experimental Setup and Evaluation Design	9
4.2	Factual Reconstruction and Decomposition	10
4.3	Commanded Position and Scale	11
4.4	Appearance-Consistent Geometry	12
4.5	Ablation Study	12
5	Conclusion	13
	References	14
	Supplementary Material	17
A	Continuous Equivariance and Scaling Proofs	17
B	Teacher–Selector Details	19
C	Implementation and Curriculum Details	20
D	Obj3D Evaluation Protocol and Extended Results	23
E	MOVi-C Protocol and Extended Diagnostics	26
F	Obj3D Ablation Protocol	29

1 Introduction

Slot-based models provide object-centric vectors for decomposition, tracking, and compositional generation. Their appeal for editing rests on a simple promise: changing an object’s representation should predictably change that object. Slot Attention [Locatello et al. \(2020\)](#) discovers such vectors through competitive binding, and SAVi [Kipf et al. \(2022\)](#) propagates them through video. Most variants still mix appearance with image-plane geometry.

ISA [Biza et al. \(2023\)](#) improves this interface by exposing an appearance vector, two-dimensional position, and scalar scale for every slot. Its relative coordinates reduce appearance’s geometric burden. Editing the exposed variables can still yield unexpected motion or resizing. Their absolute meanings may also vary across appearances: position can disagree with the rendered center, equal scales can yield different extents, and replacing appearance under fixed geometry can move or resize the object (Fig. 1).

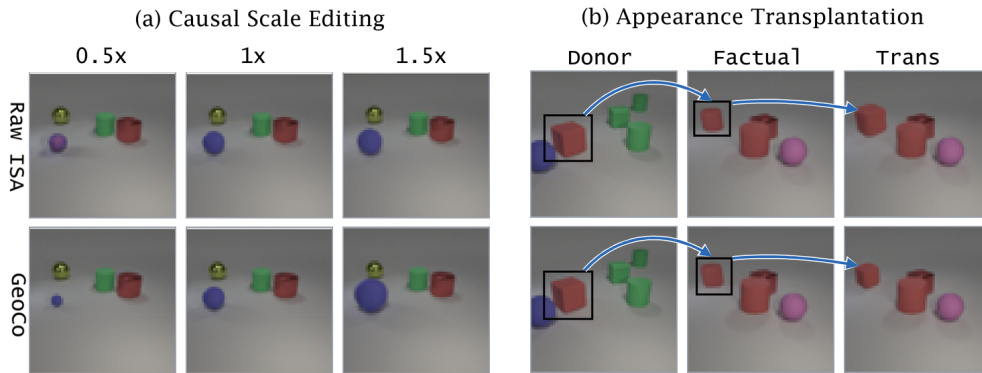


Figure 1: Raw ISA (ISA in tables) responds weakly to scale commands and can shift geometry after appearance transplantation. GeoCo-SAVi produces monotonic extent changes and preserves recipient layout.

GeoCo-SAVi addresses this gap through architecture and factual calibration. Its spatially equivariant, object-wise commanded decoder adapts SAC’s scale-conditioned sampling [Zhang et al. \(2017\)](#): explicit s_i steers every learned spatial operation along a slot’s alpha path, so changing $(\mathbf{p}_i, s_i)$ moves or resizes its support. Factual position alignment ties $\mathbf{p}_i$ to the decoded center, and normalized attention overlap discourages duplicate allocation.

Counterfactual supervision aligns geometry semantics across objects. Appearance transplantation combines donor appearance with recipient geometry, aligning decoded center and extent while retaining donor-dependent shape and compactness. Appearance may remain statistically correlated with geometry; we require predictable intervention semantics. An optional detached teacher can refine object-body ownership, and a zero-initialized temporal initializer carries calibrated slots into the next-frame ISA refinement.

Our contributions are:

- an object-wise commanded decoder with scale-steered per-slot alpha rendering, an ideal continuous covariance characterization, and explicit finite-grid limits;
- factual and transplantation objectives that calibrate position and cross-object geometry semantics while discouraging duplicate allocation; and
- optional terminal appearance readout and teacher-selector targeting reconstruction quality and more accurate object-body masks.

2 Related Work

Object-centric image and video learning. Early generative models decompose scenes through sequential inference or mixture reconstruction [Eslami et al. \(2016\)](#); [Burgess et al. \(2019\)](#); [Engelcke et al. \(2020\)](#). Slot Attention [Locatello et al. \(2020\)](#) binds visual features to exchangeable slots, and SAVi [Kipf et al. \(2022\)](#) extends this mechanism to conditional video tracking. Frozen self-supervised features improve grouping in realistic imagery [Seitzer et al. \(2023\)](#). Video methods build on self-supervision and temporal objectives [Zadaianchuk, Seitzer, and Martius \(2023\)](#); [Aydemir, Xie, and Güney \(2023\)](#); [Manasyan et al. \(2025\)](#), richer reasoning or synergistic learning [Li et al. \(2024\)](#); [Seong, Moon, and Heo \(2026\)](#); [Moon and Heo \(2026\)](#), and reconstruction curricula [Moon, Seong, and Heo \(2026\)](#). These advances target discovery and tracking; our focus is the intervention semantics of exposed geometry.

Explicit factors and controllable slots. AIR, SPACE, and SCALOR use what/where factors; ISA adds a slot-centric appearance frame [Eslami et al. \(2016\)](#); [Lin et al. \(2020b\)](#); [Jiang et al. \(2020\)](#); [Biza et al. \(2023\)](#). DISA and GOLD factor shape–texture and extrinsic–intrinsic attributes [Majellaro et al. \(2025\)](#); [Chen et al. \(2025\)](#); STAITUS and DSSA respectively separate appearance–pose and local appearance–identity [Wei, Nejjar, and Fink \(2026\)](#); [Tran et al. \(2026\)](#). SlotAug uses augmentation commands and CTRL-O language selection; foreground indicators refine support [Kim et al. \(2023\)](#); [Didolkar et al. \(2025\)](#); [Tian et al. \(2025\)](#). We test whether $(\mathbf{p}, s)$ commands remain calibrated under novel appearance–geometry combinations.

Scale equivariance. Deep scale-space models, SESN, and DISCO are scale-equivariant hierarchies [Worrall and Welling \(2019\)](#); [Sosnovik, Szmaja, and Smeulders \(2020\)](#); [Sosnovik, Moskaev, and Smeulders \(2021\)](#); SAC sets sampling dilation from an estimated scale map [Zhang et al. \(2017\)](#). We use SAC-style sampling across a slot’s alpha path: latent scale commands its footprint on a shared spatial-broadcast canvas [Watters et al. \(2019\)](#).

3 Methodology

3.1 Preliminary: Invariant Slot Attention

For a frame $\mathbf{x} \in [0, 1]^{3 \times H \times W}$ on normalized grid Ω , an encoder produces tokens $\{(\mathbf{f}_n, \mathbf{u}_n)\}_{n=1}^L$. ISA [Biza et al. \(2023\)](#) returns N slot representations: $\mathbf{a}_i \in \mathbb{R}^{D_a}$ is appearance, $\mathbf{p}_i \in [-1, 1]^2$ is image-plane position, and $s_i > 0$ is isotropic scale:

$$\mathbf{z}_i = [\mathbf{a}_i; \mathbf{p}_i; s_i]. \quad (1)$$

The shared decoder maps each slot to RGB $\mathbf{y}_i : \Omega \rightarrow [0, 1]^3$ and a raw alpha logit $\ell_i : \Omega \rightarrow \mathbb{R}$. The core contains no learned foreground, background, or empty-slot gate. Full-softmax compositing gives

$$\begin{aligned} \alpha_i(\mathbf{u}) &= \text{softmax}(\ell(\mathbf{u}))_i, \\ \hat{\mathbf{x}}(\mathbf{u}) &= \sum_i \alpha_i(\mathbf{u}) \mathbf{y}_i(\mathbf{u}). \end{aligned} \quad (2)$$

Slot-centric inference. For slot i , ISA forms slot-centric token coordinates

$$\boldsymbol{\xi}_i(\mathbf{u}_n) = \frac{\mathbf{u}_n - \mathbf{p}_i}{a_s s_i + \epsilon}, \quad (3)$$

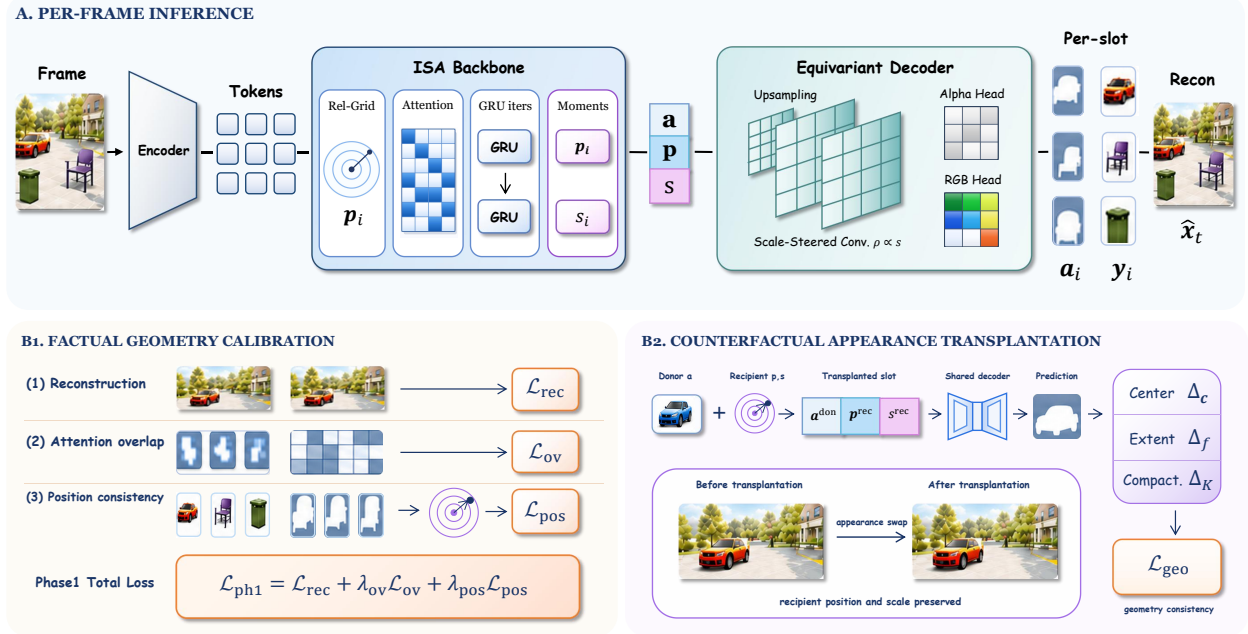


Figure 2: GeoCo-SAVi overview. *A*: ISA infers per-slot appearance and geometry; a spatially equivariant decoder renders RGB and alpha. *B1*: factual losses calibrate allocation and position. *B2*: transplants align recipient center and extent while retaining donor compactness.

where a_s is fixed. Its encoding enters attention keys and values. If $o_i(\mathbf{u}_n)$ is slot-softmax ownership, define

$$w_i(\mathbf{u}_n) = \frac{o_i(\mathbf{u}_n)}{\sum_{m=1}^L o_i(\mathbf{u}_m) + \epsilon} \quad (4)$$

ISA updates appearance recurrently and reads geometry from the normalized moments

$$\mathbf{p}_i = \sum_n w_i(\mathbf{u}_n) \mathbf{u}_n, \quad s_i = \left(\sum_n w_i(\mathbf{u}_n) \|\mathbf{u}_n - \mathbf{p}_i\|_2^2 + \epsilon \right)^{1/2}. \quad (5)$$

Slot-indexed learnable appearance and scale initializations stabilize reconstruction, while positions are initialized IID-randomly. This relaxes strict prior exchangeability; appearance transplantation remains valid. Relative coordinates reduce appearance’s geometric burden. Decoder compliance with $(\mathbf{p}_i, s_i)$ requires a commanded decoder.

Figure 2 summarizes how GeoCo-SAVi closes this gap via its core and calibration paths; Sec. 3.6 describes the omitted dataset-specific branches.

3.2 Spatially Equivariant Decoder

The desired interface is direct: changing $\mathbf{p}_i$ should translate the rendered support, while changing s_i should resize it around $\mathbf{p}_i$. Our Spatially Equivariant Decoder implements this interface as an object-wise commanded decoder. It instantiates SAC’s scale-adaptive sampling rule Zhang et al. (2017) within each slot renderer, with explicit s_i controlling that object’s sampling footprint. For its ideal continuous description, let $\mathbf{h} : \mathbb{R}^2 \rightarrow \mathbb{R}^{C_{in}}$ be a feature field and $\mathcal{K} \subset \mathbb{Z}^2$ a finite kernel support.

Its action and layerwise sampling radius are

$$\begin{aligned}(\mathcal{C}_s \mathbf{h})(\mathbf{u}) &= \sum_{\delta \in \mathcal{K}} \mathbf{W}_\delta \mathbf{h}(\mathbf{u} + \rho_\ell(s) \delta) + \mathbf{b}, \\ \rho_\ell(s) &= \Delta_\ell s / s_{\text{ref}} \kappa_\ell, \\ \kappa_\ell &= \exp(\log \kappa_{\text{max}} \tanh \psi_\ell).\end{aligned}\tag{6}$$

Here Δ_ℓ is one normalized-grid pixel, $s_{\text{ref}} > 0$ is a global gauge, and κ_ℓ is an identity-initialized bounded gain. At $s = s_{\text{ref}}$ and $\kappa_\ell = 1$, the taps match ordinary stride-one convolution.

To state the requested scale response formally, let $\mathcal{T}_{\mathbf{p},k}$ rescale canvas coordinates by $k > 0$ around $\mathbf{p}$, and let $\mathcal{D}_{\mathbf{p},k}$ apply the corresponding transformation to a feature field:

$$\begin{aligned}\mathcal{T}_{\mathbf{p},k}(\mathbf{u}) &= \mathbf{p} + k(\mathbf{u} - \mathbf{p}), \\ (\mathcal{D}_{\mathbf{p},k} \mathbf{h})(\mathbf{v}) &= \mathbf{h}(\mathcal{T}_{\mathbf{p},1/k}(\mathbf{v})).\end{aligned}\tag{7}$$

Because $\rho_\ell(ks) = k\rho_\ell(s)$, the continuous operators obey

$$\mathcal{C}_{ks} \mathcal{D}_{\mathbf{p},k} = \mathcal{D}_{\mathbf{p},k} \mathcal{C}_s.\tag{8}$$

Thus commanding ks and then filtering is equivalent, in the ideal continuous setting, to filtering at s and rescaling the resulting field around the same commanded slot center.

Proposition 1 (Continuous raw-logit covariance). Let $\ell_{\mathbf{a},\mathbf{p},s}$ be the ideal continuous per-slot alpha logit generated from the slot-centric broadcast field by pointwise maps and scale-steered layers with $\rho_\ell(ks) = k\rho_\ell(s)$. With zero coordinate stabilization and no boundary truncation, for every $\mathbf{t} \in \mathbb{R}^2$ and $k > 0$,

$$\begin{aligned}\ell_{\mathbf{a},\mathbf{p}+\mathbf{t},s}(\mathbf{u} + \mathbf{t}) &= \ell_{\mathbf{a},\mathbf{p},s}(\mathbf{u}), \\ \ell_{\mathbf{a},\mathbf{p},ks}(\mathcal{T}_{\mathbf{p},k}(\mathbf{u})) &= \ell_{\mathbf{a},\mathbf{p},s}(\mathbf{u}).\end{aligned}\tag{9}$$

Thus the ideal raw-logit path is covariant to commanded translation and scale (layerwise proof: Supplementary Material, Sec. A.1), excluding cross-slot full-softmax alpha. Finite-grid discretization, boundaries, and the bounded fixed-pixel RGB residual make equivariance approximate (Sec. 3.6).

3.3 Phase 1: Factual Geometry Calibration

Let m_i be slot i 's differentiable one-foreground (OneFG) support for a detached factual background set (Eq. (23)). For any support, $\boldsymbol{\mu}$, r , A , and $K = A/(r^2 + \epsilon)$ denote centroid, RMS radius, coverage, and compactness (Sec. 3.6).

We first optimize the per-frame reconstruction loss

$$\mathcal{L}_{\text{rec}} = \mathbb{E}_{c,\mathbf{u}} \left[(\hat{x}_c(\mathbf{u}) - x_c(\mathbf{u}))^2 \right].\tag{10}$$

After warm-up, normalized overlap calibrates slot allocation:

$$\mathcal{L}_{\text{ov}} = \mathbb{E}_{i \neq j} \left[\sum_n w_i(\mathbf{u}_n) w_j(\mathbf{u}_n) \right].\tag{11}$$

Unit spatial mass penalizes weak duplicates (uniform slot: $1/L$) without appearance-space repulsion. The position loss aligns each valid factual OneFG centroid with its slot position:

$$\mathcal{L}_{\text{pos}} = \mathbb{E}_{i \in \mathcal{V}} \left[\overline{\mathbf{H}}_{\delta_p}(\boldsymbol{\mu}(m_i) - \text{sg}(\mathbf{p}_i)) \right],\tag{12}$$

Here $\overline{\mathbf{H}}_\delta$ averages coordinatewise Huber loss and $\mathcal{V}$ indexes valid supports. Only the support path receives gradients; $(\mathbf{p}_i, s_i)$ and validity are detached (piecewise form: Supplementary Material, Sec. C.1).

3.4 Phase 2: Counterfactual Appearance Transplantation

Factual alignment cannot constrain unseen appearance–geometry combinations. We transplant donor j into recipient i as

$$\mathbf{z}_{i \leftarrow j}^{\text{cf}} = [\mathbf{a}_j^{\text{don}}; \mathbf{p}_i^{\text{rec}}; s_i^{\text{rec}}]. \quad (13)$$

For an accepted pair, m_i^{rec} and m_j^{don} are factual supports, and $m_{i \leftarrow j}^{\text{cf}}$ is decoded against the recipient’s factual background. Pair construction and detached filtering are in Supplementary Material, Sec. C.4. The geometry residuals are

$$\Delta_c^{ij} = \mu(m_{i \leftarrow j}^{\text{cf}}) - \text{sg}(\mathbf{p}_i), \quad (14)$$

$$\Delta_r^{ij} = \log r(m_{i \leftarrow j}^{\text{cf}}) - \text{sg}(\log r(m_i^{\text{rec}})), \quad (15)$$

$$\Delta_K^{ij} = \log K(m_{i \leftarrow j}^{\text{cf}}) - \text{sg}(\log K(m_j^{\text{don}})). \quad (16)$$

With detached factual confidence ω_{ij} , the objective is

$$\mathcal{L}_{\text{geo}} = \mathbb{E}_{(i,j) \sim \omega} \left[\overline{\text{H}}_{\delta_c}(\Delta_c^{ij}) + \text{H}_{\delta_r}(\Delta_r^{ij}) + \lambda_K \text{H}_{\delta_K}(\Delta_K^{ij}) \right]. \quad (17)$$

With factual evidence and recipient commands detached, $\mathcal{L}_{\text{geo}}$ updates the decoder and donor-appearance path through counterfactual alpha. Proposition 1 transports donor alpha shape under recipient geometry, ideally continuously and approximately on the finite grid. $\mathcal{L}_{\text{geo}}$ aligns center and RMS extent across appearances while preserving donor compactness, hence $A_{i \leftarrow j}^{\text{target}} = \text{sg}(K_j^{\text{don}}) \text{sg}(r_i^{\text{rec}})^2$; Supplementary Material, Sec. C.4 gives the coherence filter.

3.5 Training Objective and Curriculum

The single-frame core objective is

$$\mathcal{L}_{\text{core}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{ov}} \mathcal{L}_{\text{ov}} + \lambda_{\text{pos}} \mathcal{L}_{\text{pos}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo}} + \lambda_{\text{gate}} \mathcal{L}_{\text{gate}}. \quad (18)$$

Here $\mathcal{L}_{\text{gate}}$ is the mean squared magnitude of the optional terminal gate in Eq. (19); the selector uses its separate objective and optimizer while being jointly trained in the same Obj3D trajectory. The terminal readout, its penalty, and the selector are inactive in the reported MOVi-C configuration.

Without resetting optimization, we warm up reconstruction, ramp factual position and overlap, add counterfactual geometry, and finally train temporal initialization. The Obj3D and MOVi-C schedules, including reference-scale continuation, are in Supplementary Material, Secs. C and E, respectively. Loss weights are chosen from the observed loss and module-wise gradient magnitudes, then rounded to simple values.

3.6 Implementation Details

Optional terminal appearance readout. ISA’s final value aggregate is converted by the shared recurrent update into an appearance proposal $\tilde{\mathbf{a}}_i$. We use the bounded residual readout

$$\begin{aligned} \mathbf{a}_i^{\text{out}} &= \mathbf{a}_i + \mathbf{g}_i \odot (\tilde{\mathbf{a}}_i - \mathbf{a}_i), \\ \mathbf{g}_i &= g_{\text{read}} \tanh(\mathbf{W}_g \text{LN}[\mathbf{a}_i; \tilde{\mathbf{a}}_i - \mathbf{a}_i]). \end{aligned} \quad (19)$$

The bounded, zero-initialized gate $\mathbf{g}_i$ starts from the identity. The returned $\mathbf{a}_i^{\text{out}}$ enters both decoder branches and is subsequently abbreviated as $\mathbf{a}_i$. It improves RGB reconstruction without directly updating $\mathbf{p}_i$ or s_i .

Discrete decoder realization. Following spatial broadcast [Watters et al. \(2019\)](#), slot i induces

$$\mathbf{h}_i^{(0)}(\mathbf{u}) = \text{Broadcast}(\mathbf{a}_i) + \text{PE}\left(\frac{\mathbf{u} - \mathbf{p}_i}{a_s s_i + \epsilon}\right). \quad (20)$$

A learned pointwise embedding PE is injected once, after which progressive bilinear synthesis alternates with bilinearly sampled scale-steered convolutions. The pointwise alpha head is

$$\ell_i(\mathbf{u}) = \mathbf{w}_\alpha^\top \mathbf{h}_i^{\text{eq}}(\mathbf{u}) + b_\alpha, \quad (21)$$

so every learned spatial operation on its path is scale-steered. RGB uses a pointwise fusion of $\mathbf{h}_i^{\text{eq}}$ with broadcast appearance to form $\mathbf{h}_i^{\text{rgb}}$, followed by

$$\mathbf{y}_i(\mathbf{u}) = \sigma\left(\mathbf{y}_i^{\text{eq}}(\mathbf{u}) + \eta_{\max} \tanh\left[\mathcal{C}_{3 \times 3}^{\text{micro}} \mathbf{h}_i^{\text{rgb}}(\mathbf{u})\right]\right), \quad (22)$$

where the fixed-pixel micro convolution is zero-initialized and the residual is bounded by $\eta_{\max}$. We enable it for Obj3D RGB; being bounded, it refines appearance while leaving the alpha and geometry paths untouched. Here $\mathbf{h}_i^{\text{eq}}$ is the final equivariant feature field and $\mathbf{y}_i^{\text{eq}}$ its scale-steered RGB-logit readout. Sampling, padding, clipping, aliasing, and cross-slot competition are the finite-grid approximation boundaries. Supplementary Material, Secs. [C.2–C.3](#) give the discrete operator and topology.

Differentiable geometry support. Geometry training uses the differentiable logits of a detached factual background set $\mathcal{B}$:

$$\ell_{\mathcal{B}}(\mathbf{u}) = \text{logsumexp}_{j \in \mathcal{B}} \ell_j(\mathbf{u}), \quad m_i(\mathbf{u}) = \sigma(\gamma[\ell_i(\mathbf{u}) - \ell_{\mathcal{B}}(\mathbf{u})]), \quad \gamma = 2. \quad (23)$$

This sharpened OneFG mask removes other foreground competitors. Invisible geometry remains outside the proxy. For nonnegative m , define

$$\begin{aligned} M_m &= \sum_{\mathbf{u}} m(\mathbf{u}), \quad \boldsymbol{\mu}(m) = (M_m + \epsilon)^{-1} \sum_{\mathbf{u}} m(\mathbf{u}) \mathbf{u}, \\ r(m)^2 &= (M_m + \epsilon)^{-1} \sum_{\mathbf{u}} m(\mathbf{u}) \|\mathbf{u} - \boldsymbol{\mu}(m)\|_2^2 + \epsilon, \\ A(m) &= \mathbb{E}_{\mathbf{u}}[m(\mathbf{u})], \quad K(m) = (r(m)^2 + \epsilon)^{-1} A(m). \end{aligned} \quad (24)$$

Under uniform scaling, r is first-order, A second-order, and K invariant (Supplementary Material, Sec. [A.2](#)). OneFG is used only as a differentiable training proxy; evaluation uses hard full-softmax ownership.

Optional selective alpha ownership. A detached training-only teacher improves body support by labeling foreground-owned pixels as transfer, protect, or unknown from RGB reconstruction and renderer evidence. A lightweight selector is jointly trained with a disjoint optimizer to predict these labels. At deployment it uses only detached internal ownership, logit, slot-feature, and renderer-difference evidence. For an accepted pixel, let $d_i = \ell_i - \ell_{\text{bg}}$ and $d_{\text{alt}} = \ell_{\text{alt}} - \ell_{\text{bg}}$ denote the owner and strongest alternative foreground gaps. The minimum symmetric transfer needed for background to win by margin m_{bg} is

$$\Delta_{\text{req}}(x) = \max\{0, (d_i + m_{\text{bg}})/2, d_{\text{alt}} + m_{\text{bg}}\}. \quad (25)$$

The accepted-pixel update is

$$\Delta_{\text{tr}} = \min(\Delta_{\text{req}}, \Delta_{\text{max}}), \quad \ell'_i = \ell_i - \Delta_{\text{tr}}, \quad \ell'_{\text{bg}} = \ell_{\text{bg}} + \Delta_{\text{tr}}. \quad (26)$$

The action is zero elsewhere. Its threshold and cap are fixed on a calibration panel drawn only from the training split. Supplementary Material, Secs. [A.3–B](#) give the action, minimality proof, teacher, and feature definitions.

Temporal initializer. A STATM-style Li et al. (2024) Transformer maps preceding raw-scale slot states to zero-initialized residuals in appearance, position, and log-scale. It starts from exact carry-forward and is trained by next-frame reconstruction through current-frame ISA refinement:

$$(\delta \mathbf{a}, \delta \mathbf{p}, \delta \ell)_{t,i} = \mathbf{W}_{\text{init}}(\mathcal{F}_{\Theta}(\mathbf{Z}_{<t}))_i, \quad \mathbf{p}_{t,i}^{(0)} = \text{clip}(\mathbf{p}_{t-1,i} + \delta \mathbf{p}_{t,i}), \quad s_{t,i}^{(0)} = \text{clip}(s_{t-1,i} \exp(\delta \ell_{t,i})). \quad (27)$$

Supplementary Material, Sec. C.6 gives its training semantics.

4 Experiments

We evaluate three questions: (1) whether GeoCo-SAVi preserves factual reconstruction and decomposition quality; (2) whether position and scale behave as commands; and (3) whether their meaning remains stable across appearances. Controlled single-frame Obj3D Lin et al. (2020a) isolates these effects, while MOVi-C Greff et al. (2022) tests them on complex video objects.

4.1 Experimental Setup and Evaluation Design

Shared protocol and three support roles. All experiments use one NVIDIA A800 80GB GPU. *Training support* consists solely of model-produced decoder and attention supports, including the differentiable OneFG proxy in Eq. (23); neither GT nor engineered evaluation labels enter optimization. *Reference support* locks pre-edit identity using input-only RGB cores on Obj3D and official GT masks on MOVi-C, without rematching. *Measurement support* is each method’s argmax full-softmax decoded mask, so its moments evaluate control of rendered support.

Controlled Obj3D. Obj3D contains 64×64 scenes with moving colored solids. ISA, GeoCo-SAVi, and same-protocol SlotAug Kim et al. (2023) use six 64-dimensional appearance states and train from scratch for 70k updates with matched capacity and sample exposure. ISA uses reconstruction alone; GeoCo-SAVi follows Sec. 3.5; SlotAug retains its published objective and commands, with command normalization delayed to a later hidden layer to keep the instruction identifiable. Four inference initializations cover the validation set. Frozen locking yields 697 common object locks and 249 complete eight-appearance groups. Shared-lock scoring fixes the object set across methods; feasibility is applied only after locking and never inspects edit accuracy.

PSNR and attention overlap use the full validation population. On shared locks, p -centroid error tests factual alignment, translation error tests position commands, fixed- s variation and transfer drift test appearance leakage, and log-slopes and monotonicity test scale response. Centroid and drift are normalized-grid L_2 distances; radius is grid RMS, coverage an area fraction, and translation is normalized by the 64×64 diagonal. Supplementary Material, Sec. D gives full population, unit, lock, and feasibility definitions.

MOVi-C. We use all 24 frames of the official 250-video validation set. All methods reconstruct 64×64 RGB from frozen DINOv2 features Oquab et al. (2024). GeoCo-SAVi uses 11 slots, 64-dimensional appearance, and a 100k curriculum: two-frame clips through 70k, then four-frame clips with a zero-residual temporal initializer. Its optional terminal appearance readout is disabled because PSNR is secondary here. Same-protocol SlotContrast Manasyan et al. (2025) and STAITUS Wei, Nejjar, and Fink (2026) match capacity, updates, sample exposure, and common settings. SlotContrast replaces feature prediction with RGB reconstruction while retaining temporal similarity for a fair PSNR comparison. Positive STAITUS sparsity collapses the active-slot count here, so we use zero, its most favorable value; remaining method-specific hyperparameters

FACTUAL DECOMPOSITION AND STABILITY

Metric	ISA	SlotAug	Ours
PSNR (dB) $\uparrow$	44.145	43.216 $\downarrow$ 0.929 dB	<u>44.046</u> $\downarrow$ 0.099 dB
p -centroid error $\downarrow$	0.1084	<u>0.0482</u> $\downarrow$ 55.5%	0.0208 $\downarrow$ 80.8%
Attention overlap $\downarrow$	0.0148	<u>0.0124</u> $\downarrow$ 16.2%	0.0007 $\downarrow$ 95.3%
Fixed- s radius std. $\downarrow$	0.0198	<u>0.0178</u> $\downarrow$ 10.1%	0.0089 $\downarrow$ 55.1%
Fixed- s coverage std. $\downarrow$	0.0077	<u>0.0064</u> $\downarrow$ 16.9%	0.0026 $\downarrow$ 66.2%

COUNTERFACTUAL EDITING CAPABILITY

Metric	ISA	SlotAug	Ours
Translation error $\downarrow$	0.0071	<u>0.0044</u> $\downarrow$ 38.0%	0.0037 $\downarrow$ 47.9%
Radius log-slope	0.0420	<u>0.8950</u> $\uparrow$ 89.0%	1.0160 $\uparrow$ 98.3%
Coverage log-slope	0.1110	<u>1.8000</u> $\uparrow$ 89.4%	1.9570 $\uparrow$ 97.7%
Radius monotonicity $\uparrow$	23.3%	<u>70.3%</u> $\uparrow$ 47.0 pp	100.0% $\uparrow$ 76.7 pp
Coverage monotonicity $\uparrow$	30.3%	<u>70.3%</u> $\uparrow$ 40.0 pp	100.0% $\uparrow$ 69.7 pp
Transfer centroid drift $\downarrow$	0.0596	<u>0.0516</u> $\downarrow$ 13.4%	0.0158 $\downarrow$ 73.5%

Table 1: Single-frame Obj3D results. PSNR and attention overlap use the full validation population; p -centroid error starts from 697 common reference locks, fixed- s statistics use 249 complete groups, and intervention rows use the predeclared task-specific feasible records. Best values are bold and second-best values are underlined. Colored deltas are relative to ISA (green/red: better/worse); slope percentages reduce $|\beta - \beta^*|$.

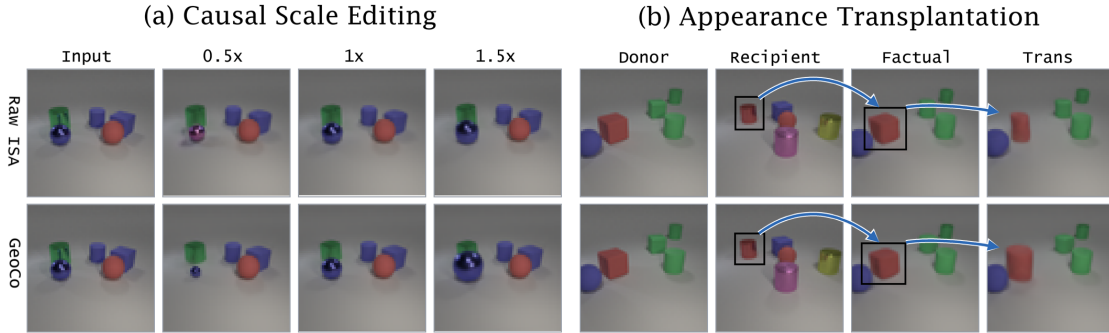


Figure 3: Counterfactual control on complementary Obj3D examples. GeoCo-SAVi produces a monotonic extent response to scale commands (*left*) and places donor appearance at recipient geometry (*right*); Raw ISA is comparatively scale-insensitive and its transplanted support drifts.

are published values. These compute-matched results support within-table comparison under the shared configuration.

Factual metrics use all 250 videos without filtering. Intervention analysis starts from 1,543 GT-defined candidate tracks and retains 178 common pre-edit locks, fixing the same output-independent identities across methods. Edited geometry uses decoded support without GT rematching. Supplementary Material, Sec. E gives implementation and lock details.

4.2 Factual Reconstruction and Decomposition

Per-example PSNR uses $10 \log_{10}(1/\text{MSE})$ for RGB in $[0, 1]$. On MOVi-C, full-softmax alpha is resized to GT resolution before slotwise argmax; FG-ARI, ARI, Hungarian mIoU, and mBO are then

Method	PSNR (dB) $\uparrow$	FG-ARI $\uparrow$	ARI $\uparrow$	Hungarian mIoU $\uparrow$	mBO $\uparrow$
SlotContrast	16.467	0.1034	0.0277	0.0504	0.0599
STAITUS	<u>19.393</u>	<u>0.1550</u>	<u>0.1114</u>	<u>0.1242</u>	<u>0.1307</u>
GeoCo-SAVi	22.233	0.1935	0.4610	0.2451	0.2507

Table 2: MOVi-C factual reconstruction and instance grouping on the complete 250-video validation set. Each video has equal weight. Best values are bold and second-best values are underlined.

FACTUAL DECOMPOSITION AND STABILITY			
Metric	GeoCo-SAVi	STAITUS	SlotContrast
p -centroid error $\downarrow$	0.0722 $\downarrow$ 40.4%	<u>0.1212</u>	N/A
Attention overlap $\downarrow$	<u>0.0011</u> $\uparrow$ 37.5%	0.0008	0.0018
Fixed- s radius std. $\downarrow$	0.0117 $\downarrow$ 52.0%	<u>0.0244</u>	N/A
Fixed- s coverage std. $\downarrow$	0.0080 $\downarrow$ 18.4%	<u>0.0098</u>	N/A

COUNTERFACTUAL EDITING CAPABILITY			
Metric	GeoCo-SAVi	STAITUS	SlotContrast
Translation error $\downarrow$	0.0041 $\downarrow$ 73.2%	<u>0.0153</u>	N/A
Radius log-slope	0.9570 $\uparrow$ 84.0%	<u>0.7310</u>	N/A
Coverage log-slope	1.8300 $\uparrow$ 77.2%	<u>1.2550</u>	N/A
Radius monotonicity $\uparrow$	100.0%	100.0%	N/A
Coverage monotonicity $\uparrow$	100.0% $\uparrow$ 7.7 pp	<u>92.3%</u>	N/A
Transfer centroid drift $\downarrow$	0.0265 $\downarrow$ 45.1%	<u>0.0483</u>	N/A

Table 3: MOVi-C geometry and intervention metrics starting from 178 common pre-edit track-slot locks; task-specific output-feasibility rules are applied after locking. Attention overlap uses every frame of all 250 validation videos. Radius and coverage log-slope targets are 1 and 2. Best values are bold and second-best values are underlined. Colored deltas compare GeoCo-SAVi with STAITUS (green/red: better/worse); slope percentages reduce $|\beta - \beta^*|$.

computed per video and averaged with equal video weight. On Obj3D, GeoCo-SAVi matches ISA reconstruction (Table 1) while reducing attention overlap by 95.3%, indicating cleaner allocation without sacrificing RGB fidelity. On MOVi-C, Table 2 evaluates all 250 videos without reference-lock filtering: GeoCo-SAVi leads both same-protocol references in PSNR and all four instance-grouping metrics. Thus the added geometric structure preserves controlled-scene reconstruction and improves factual decomposition in complex video.

4.3 Commanded Position and Scale

Position. On Obj3D, GeoCo-SAVi reduces p -centroid error by 80.8% relative to ISA, aligning the explicit coordinate with decoded support. Under four ± 6 -pixel commands, translation error is 0.0037, versus 0.0044 for SlotAug and 0.0071 for ISA; GeoCo-SAVi and ISA update $\mathbf{p}_i$ directly, while SlotAug uses its native instruction. On MOVi-C, the 178-lock analysis uses GT only to establish pre-edit identity, then scores locked decoded masks without re-inference or rematching. GeoCo-SAVi reduces p -centroid error from 0.1212 to 0.0722 and translation error from 0.0153 to 0.0041 relative to STAITUS (Table 3). SlotContrast lacks a direct $(\mathbf{a}, \mathbf{p}, s)$ interface, so its counterfactual entries are N/A.

Scale. We use scale factors 0.5, 0.75, 1, 1.25, and 1.5. For each $x \in \{r, A\}$, we regress $\log[x(k)/x(1)]$ on $\log k$ and report the fitted slope β_x . Against ideal slopes (1, 2), GeoCo-SAVi obtains (1.016, 1.957) on Obj3D and (0.957, 1.830) on MOVi-C, with 100.0% monotonicity for both measures. ISA is largely scale-insensitive, while STAITUS reaches (0.731, 1.255) on MOVi-C (Tables 1 and 3; Fig. 3). These finite-grid responses match the trend predicted by Proposition 1 without extending its guarantee to cross-slot softmax alpha. The post-edit gate checks only canvas feasibility, never edit accuracy, so all means remain conditional on feasible support (Supplementary Material, Secs. D–E).

4.4 Appearance-Consistent Geometry

Holding recipient geometry fixed while replacing appearance reduces Obj3D transfer centroid drift to 0.0158, versus 0.0596 for ISA and 0.0516 for SlotAug. Across 249 groups assigning eight appearances the same $(\mathbf{p}, s)$, GeoCo-SAVi reduces hard-radius and coverage variation by 55.1% and 66.2%; these within-group statistics isolate appearance leakage at fixed recipient geometry. The same pattern holds on MOVi-C: relative to STAITUS, GeoCo-SAVi cuts transfer drift by 45.1% and approximately halves fixed- s radius variation (Table 3). Together with Fig. 3, these results show that the commands retain their meaning across donor appearances.

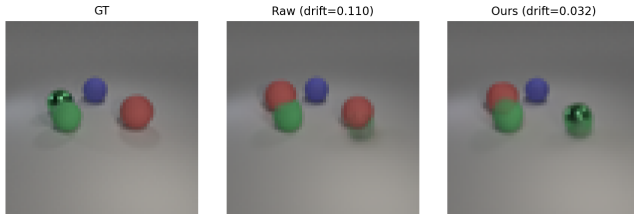


Figure 4: Factual calibration reduces appearance-swap centroid drift: uncalibrated Raw ISA displaces the replaced object, whereas the calibrated diagnostic preserves its recipient location. This 40k diagnostic follows a separate trajectory using reconstruction, position, and normalized-overlap losses.

STAITUS’s attention overlap is computed only on active slots, consistent with its paper, whereas the other methods include empty slots with uniform attention; this convention favors lower STAITUS values. Full lock and validity rules are in Supplementary Material, Secs. D–E.

4.5 Ablation Study

Four controls isolate the roles of decoder equivariance, allocation regularization, factual position calibration, and counterfactual geometry calibration.

Spatial equivariance. We replace the spatially equivariant decoder by a conventional decoder and calibrate scale against factual spread. At $2\times$ scale, spread is nearly correct. Coverage and density collapse (Table 4). The resulting bell-shaped support shows that scale authority requires control of coverage and density as well as spread.

Attention allocation. Without $\mathcal{L}_{ov}$, factual attention overlap rises from 0.0007 to 0.0027, indicating that spatial normalization discourages duplicate attention without a foreground/background gate.

Factual position calibration. Without $\mathcal{L}_{pos}$, final $\mathcal{L}_{geo}$ rises from 0.0018 to 0.0031. Recipient-center targets are therefore unreliable until factual position is aligned with the support centroid.

Counterfactual geometry. Without $\mathcal{L}_{geo}$, transfer centroid drift increases from 0.0158 to 0.0298 and fixed- s radius std. from 0.0089 to 0.0095, isolating $\mathcal{L}_{geo}$ ’s role in semantic alignment under transplantation. This ablation is sensitive to the learned position semantics.

Diagnostic	Ablated	Ref. / Ideal
<i>Conventional decoder, $2\times$ scale</i>		
Spread ratio R_r	1.978	2
Soft coverage R_{soft}	0.708	4
γ -2 coverage $R_{\gamma=2}$	0.0385	4
Soft density R_K	0.178	1
<i>Without $\mathcal{L}_{\text{ov}}$</i>		
Attention overlap $\downarrow$	0.0027	0.0007
<i>Without $\mathcal{L}_{\text{pos}}$</i>		
Final $\mathcal{L}_{\text{geo}} \downarrow$	0.0031	0.0018
<i>Without $\mathcal{L}_{\text{geo}}$</i>		
Transfer centroid drift $\downarrow$	0.0298	0.0158
Fixed- s radius std. $\downarrow$	0.0095	0.0089

Table 4: Obj3D mechanism ablations. *Ref. / Ideal* is the full-model value or analytic target; see Supplementary Material, Sec. F.

5 Conclusion

Summary. GeoCo-SAVi makes explicit position and scale effective commands with meaning aligned across appearances. Its spatially equivariant decoder makes edits move or resize support; factual calibration aligns commands with decoded geometry, and appearance transplantation aligns them across objects. GeoCo-SAVi matches ISA reconstruction and improves Obj3D geometry and editing plus MOVi-C reconstruction, grouping, and fixed-lock control over same-protocol references. The learned geometry supports controlled intervention beyond post-hoc prediction, making exposed geometry command the decoded layout.

Contributions. (1) *Spatially equivariant decoding.* Our object-wise commanded decoder uses explicit $\mathbf{p}$ for position and SAC-style scale-conditioned sampling throughout each alpha path for s -controlled extent. We characterize ideal continuous raw-logit covariance and its finite-grid limits. (2) *Calibration by intervention.* Factual objectives align support centroids and allocation; transplantation uses recipient geometry for layout while retaining donor shape and compactness, aligning center and extent across appearances. (3) *Optional optimization components.* These target better reconstruction and more accurate object-body masks.

Limitations. (1) *Guarantee.* Commutation covers the ideal continuous per-slot raw-logit operator on $\mathbb{R}^2$; discretization, padding, clipping, stabilization, and cross-slot softmax make finite-grid final-alpha equivariance approximate. (2) *Geometry.* Scale is isotropic, cannot express anisotropy or orientation, and decoded alpha need not equal the physical object body. (3) *Occlusion.* Depth is not explicit, so appearance still mediates occlusion order, while partial occlusion can impair center and extent recovery. (4) *Operating regime.* Conservative pre-edit locks and canvas-feasible supports measure control given stable identity. Dataset-wide discovery and unconditional edit success remain outside this evaluation. (5) *Breadth.* Full video-training cost precluded exhaustive sweeps across settings; broader evaluation would strengthen the evidence.

Future work. (1) *Broader validation.* Validate on more complex datasets and stronger configurations across wider seed–hyperparameter ranges. (2) *Depth-aware geometry.* Use 3D depth and compositing to decouple occlusion order from appearance and recover position/scale under partial visibility. (3) *World-model trajectories.* Recover real-coordinate object trajectories for world models using calibrated image-plane, camera, and scene geometry.

References

- Aydemir, G.; Xie, W.; and Güney, F. 2023. Self-supervised Object-Centric Learning for Videos. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Ba, J. L.; Kiros, J. R.; and Hinton, G. E. 2016. Layer Normalization. *arXiv preprint arXiv:1607.06450*.
- Biza, O.; van Steenkiste, S.; Sajjadi, M. S. M.; Elsayed, G. F.; Mahendran, A.; and Kipf, T. 2023. Invariant Slot Attention: Object Discovery with Slot-Centric Reference Frames. In *International Conference on Machine Learning (ICML)*.
- Burgess, C. P.; Matthey, L.; Watters, N.; Kabra, R.; Higgins, I.; Botvinick, M.; and Lerchner, A. 2019. MONet: Unsupervised Scene Decomposition and Representation. *arXiv preprint arXiv:1901.11390*.
- Chen, T.; Huang, Y.; Shen, Z.; Huang, J.; Li, B.; and Xue, X. 2025. Learning Global Object-Centric Representations via Disentangled Slot Attention. *Machine Learning*, 114(2): 40.
- Cho, K.; van Merriënboer, B.; Gulcehre, C.; Bahdanau, D.; Bougares, F.; Schwenk, H.; and Bengio, Y. 2014. Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Didolkar, A.; Zadaianchuk, A.; Awal, R.; Seitzer, M.; Gavves, E.; and Agrawal, A. 2025. CTRL-O: Language-Controllable Object-Centric Visual Representation Learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 29523–29533.
- Engelcke, M.; Kosiorek, A. R.; Parker Jones, O.; and Posner, I. 2020. GENESIS: Generative Scene Inference and Sampling with Object-Centric Latent Representations. In *International Conference on Learning Representations (ICLR)*.
- Eslami, S. M. A.; Heess, N.; Weber, T.; Tassa, Y.; Szepesvari, D.; Kavukcuoglu, K.; and Hinton, G. E. 2016. Attend, Infer, Repeat: Fast Scene Understanding with Generative Models. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Greff, K.; Belletti, F.; Beyer, L.; Doersch, C.; et al. 2022. Kubric: A scalable dataset generator. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Jiang, J.; Janghorbani, S.; de Melo, G.; and Ahn, S. 2020. SCALOR: Generative World Models with Scalable Object Representations. In *International Conference on Learning Representations (ICLR)*.
- Kim, J.; Choi, J.; Kang, J.; Lee, C.; Choi, H.-J.; and Kim, S. J. 2023. Leveraging Image Augmentation for Object Manipulation: Towards Interpretable Controllability in Object-Centric Learning. *arXiv preprint arXiv:2310.08929*.
- Kipf, T.; Elsayed, G. F.; Mahendran, A.; Stone, A.; Sabour, S.; Heigold, G.; Jonschkowski, R.; Dosovitskiy, A.; and Greff, K. 2022. Conditional Object-Centric Learning from Video. In *International Conference on Learning Representations (ICLR)*.
- Kuhn, H. W. 1955. The Hungarian Method for the Assignment Problem. *Naval Research Logistics Quarterly*, 2(1–2): 83–97.

- Li, J.; Ren, P.; Liu, Y.; and Sun, H. 2024. Reasoning-Enhanced Object-Centric Learning for Videos. *arXiv preprint arXiv:2403.15245*.
- Lin, Z.; Wu, Y.-F.; Peri, S.; Fu, B.; Jiang, J.; and Ahn, S. 2020a. Improving Generative Imagination in Object-Centric World Models. In *International Conference on Machine Learning (ICML)*.
- Lin, Z.; Wu, Y.-F.; Peri, S. V.; Sun, W.; Singh, G.; Deng, F.; Jiang, J.; and Ahn, S. 2020b. SPACE: Unsupervised Object-Oriented Scene Representation via Spatial Attention and Decomposition. In *International Conference on Learning Representations (ICLR)*.
- Locatello, F.; Weissenborn, D.; Unterthiner, T.; Mahendran, A.; Heigold, G.; Uszkoreit, J.; Dosovitskiy, A.; and Kipf, T. 2020. Object-Centric Learning with Slot Attention. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Loshchilov, I.; and Hutter, F. 2019. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations (ICLR)*.
- Majellaro, R.; Collu, J.; Plaat, A.; and Moerland, T. M. 2025. Explicitly Disentangled Representations in Object-Centric Learning. *Transactions on Machine Learning Research*.
- Manasyan, A.; Seitzer, M.; Radovic, F.; Martius, G.; and Zadaianchuk, A. 2025. Temporally Consistent Object-Centric Learning by Contrasting Slots. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 5401–5411.
- Moon, W.; and Heo, J.-P. 2026. Selective Synergistic Learning for Video Object-Centric Learning. *arXiv preprint arXiv:2606.15527*.
- Moon, W.; Seong, H. S.; and Heo, J.-P. 2026. Reconstruction-Guided Slot Curriculum: Addressing Object Over-Fragmentation in Video Object-Centric Learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 25001–25010.
- Oquab, M.; Darcet, T.; Moutakanni, T.; Vo, H.; Szafraniec, M.; Khalidov, V.; Fernandez, P.; Haziza, D.; Massa, F.; El-Nouby, A.; Assran, M.; Ballas, N.; Galuba, W.; Howes, R.; Huang, P.-Y.; Li, S.-W.; Misra, I.; Rabbat, M.; Sharma, V.; Synnaeve, G.; Xu, H.; Jegou, H.; Mairal, J.; Labatut, P.; Joulin, A.; and Bojanowski, P. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research*.
- Seitzer, M.; Horn, M.; Zadaianchuk, A.; Zietlow, D.; Xiao, T.; Simon-Gabriel, C.-J.; He, T.; Zhang, Z.; Schölkopf, B.; Brox, T.; and Locatello, F. 2023. Bridging the Gap to Real-World Object-Centric Learning. In *International Conference on Learning Representations (ICLR)*.
- Seong, H. S.; Moon, W.; and Heo, J.-P. 2026. From Vicious to Virtuous Cycles: Synergistic Representation Learning for Unsupervised Video Object-Centric Learning. In *International Conference on Learning Representations (ICLR)*.
- Sosnovik, I.; Moskalev, A.; and Smeulders, A. W. M. 2021. DISCO: Accurate Discrete Scale Convolutions. In *British Machine Vision Conference (BMVC)*.
- Sosnovik, I.; Szmaja, M.; and Smeulders, A. W. M. 2020. Scale-Equivariant Steerable Networks. In *International Conference on Learning Representations (ICLR)*.
- Tian, P.; Yang, S.; Yu, H.; and Kot, A. C. 2025. Pay Attention to the Foreground in Object-Centric Learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 30281–30290.
- Tran, S.; Nguyen, D.; Vo, H.; Vo, K.; and Le, N. 2026. Dual-State Slot Attention: Decoupling Appearance and Identity for Video Object-Centric Learning. *arXiv preprint arXiv:2606.12601*.

- Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention Is All You Need. *arXiv preprint arXiv:1706.03762*.
- Watters, N.; Matthey, L.; Burgess, C. P.; and Lerchner, A. 2019. Spatial Broadcast Decoder: A Simple Architecture for Learning Disentangled Representations in VAEs. *arXiv preprint arXiv:1901.07017*.
- Wei, A.; Nejjar, I.; and Fink, O. 2026. Rethinking Object-Centric Representations for Video Dynamics Modeling. *arXiv preprint arXiv:2606.23436*.
- Worrall, D. E.; and Welling, M. 2019. Deep Scale-spaces: Equivariance Over Scale. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 7364–7376.
- Zadaianchuk, A.; Seitzer, M.; and Martius, G. 2023. Object-Centric Learning for Real-World Videos by Predicting Temporal Feature Similarities. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Zhang, R.; Tang, S.; Zhang, Y.; Li, J.; and Yan, S. 2017. Scale-Adaptive Convolutions for Scene Parsing. In *IEEE International Conference on Computer Vision (ICCV)*.

Supplementary Material

A Continuous Equivariance and Scaling Proofs

A.1 Scale-Steered Convolution

Lemma 1 (Layer commutation). Let $\mathcal{C}_s$ be the continuous scale-steered layer in Eq. (6), acting on fields over $\mathbb{R}^2$, and assume $\rho_\ell(k s) = k \rho_\ell(s)$ for every $k > 0$. Define translation by $(\mathcal{D}_{\mathbf{t}} \mathbf{h})(\mathbf{v}) = \mathbf{h}(\mathbf{v} - \mathbf{t})$. With no boundary truncation,

$$\mathcal{C}_{ks} \mathcal{D}_{\mathbf{p},k} = \mathcal{D}_{\mathbf{p},k} \mathcal{C}_s, \quad \mathcal{C}_s \mathcal{D}_{\mathbf{t}} = \mathcal{D}_{\mathbf{t}} \mathcal{C}_s. \quad (\text{A.1})$$

Every pointwise affine map or nonlinearity commutes with both actions as well.

Proof. For scale covariance, evaluate the left-hand side at $\mathcal{T}_{\mathbf{p},k}(\mathbf{u})$:

$$\begin{aligned} & (\mathcal{C}_{ks} \mathcal{D}_{\mathbf{p},k} \mathbf{h})(\mathcal{T}_{\mathbf{p},k}(\mathbf{u})) \\ &= \sum_{\delta \in \mathcal{K}} \mathbf{W}_\delta (\mathcal{D}_{\mathbf{p},k} \mathbf{h})(\mathcal{T}_{\mathbf{p},k}(\mathbf{u}) + \rho_\ell(k s) \delta) + \mathbf{b} \\ &= \sum_{\delta \in \mathcal{K}} \mathbf{W}_\delta \mathbf{h} \left(\mathbf{u} + \frac{\rho_\ell(k s)}{k} \delta \right) + \mathbf{b} \\ &= \sum_{\delta \in \mathcal{K}} \mathbf{W}_\delta \mathbf{h}(\mathbf{u} + \rho_\ell(s) \delta) + \mathbf{b} = (\mathcal{C}_s \mathbf{h})(\mathbf{u}). \end{aligned} \quad (\text{A.2})$$

The third line applies $\mathcal{T}_{\mathbf{p},1/k}$ to the sampling coordinate, and the fourth uses $\rho_\ell(k s) = k \rho_\ell(s)$. By the definition of $\mathcal{D}_{\mathbf{p},k}$, the final expression equals $(\mathcal{D}_{\mathbf{p},k} \mathcal{C}_s \mathbf{h})(\mathcal{T}_{\mathbf{p},k}(\mathbf{u}))$. Translation follows from

$$(\mathcal{C}_s \mathcal{D}_{\mathbf{t}} \mathbf{h})(\mathbf{u} + \mathbf{t}) = \sum_{\delta \in \mathcal{K}} \mathbf{W}_\delta \mathbf{h}(\mathbf{u} + \rho_\ell(s) \delta) + \mathbf{b} = (\mathcal{C}_s \mathbf{h})(\mathbf{u}). \quad (\text{A.3})$$

Finally, for a pointwise map ϕ , $\phi(\mathcal{D} \mathbf{h})(\mathbf{v}) = \phi(\mathbf{h}(\mathcal{D}^{-1} \mathbf{v})) = (\mathcal{D} \phi(\mathbf{h}))(\mathbf{v})$. $\square$

Theorem 1 (Per-slot raw-logit covariance). Set the coordinate stabilizer to zero. Let the initial per-slot field be

$$\mathbf{h}_{\mathbf{a},\mathbf{p},s}^{(0)}(\mathbf{u}) = \text{Broadcast}(\mathbf{a}) + \text{PE} \left(\frac{\mathbf{u} - \mathbf{p}}{a_s s} \right), \quad (\text{A.4})$$

and let $\ell_{\mathbf{a},\mathbf{p},s}$ be obtained from this field by any finite composition of pointwise maps and scale-steered layers satisfying the assumptions of Lemma 1, followed by a pointwise alpha-logit head. Then

$$\begin{aligned} \ell_{\mathbf{a},\mathbf{p}+\mathbf{t},s}(\mathbf{u} + \mathbf{t}) &= \ell_{\mathbf{a},\mathbf{p},s}(\mathbf{u}), \\ \ell_{\mathbf{a},\mathbf{p},ks}(\mathcal{T}_{\mathbf{p},k}(\mathbf{u})) &= \ell_{\mathbf{a},\mathbf{p},s}(\mathbf{u}) \end{aligned} \quad (\text{A.5})$$

for every $\mathbf{t} \in \mathbb{R}^2$ and $k > 0$.

Proof. The broadcast is translation covariant because

$$\frac{(\mathbf{u} + \mathbf{t}) - (\mathbf{p} + \mathbf{t})}{a_s s} = \frac{\mathbf{u} - \mathbf{p}}{a_s s},$$

and scale covariant because

$$\frac{\mathcal{T}_{\mathbf{p},k}(\mathbf{u}) - \mathbf{p}}{a_s k s} = \frac{\mathbf{u} - \mathbf{p}}{a_s s}.$$

Thus Eq. (A.5) holds at layer zero. Lemma 1 preserves the corresponding identity after each scale-steered or pointwise layer. Induction over the finite decoder path, including the pointwise alpha head, proves the result and hence Proposition 1 in the main paper. $\square$

Remark 1 (Scope). The theorem is exact only for the ideal continuous per-slot raw-logit operator. Finite-grid interpolation, padding, clipping, nonzero numerical stabilization, and boundary truncation are excluded. If all slot logits, including background logits, undergo the same global action, pointwise softmax also covaries. Under an intervention that changes only one slot, the other logits remain fixed and cross-slot full-softmax competition generally prevents exact alpha equivariance.

A.2 Support Moments

Assume an isolated continuous support obeys

$$m_{ks}(\mathcal{T}_{\mathbf{p},k}(\mathbf{u})) = m_s(\mathbf{u}). \quad (\text{A.6})$$

The substitution $\mathbf{v} = \mathcal{T}_{\mathbf{p},k}(\mathbf{u})$ has area element $d\mathbf{v} = k^2 d\mathbf{u}$. It follows that

$$\boldsymbol{\mu}(ks) = \mathbf{p} + k[\boldsymbol{\mu}(s) - \mathbf{p}], \quad r(ks) = kr(s), \quad (\text{A.7})$$

and $A(ks) = k^2 A(s)$ whenever the support is not clipped by the canvas. Consequently,

$$K(ks) = \frac{k^2 A(s)}{k^2 r(s)^2} = K(s). \quad (\text{A.8})$$

Thus Eq. (17) jointly aligns radius and compactness.

A.3 Minimal Owner-to-Background Transfer

Main-paper Eqs. (25)–(26) define the required and capped symmetric transfers; the action is zero outside accepted pixels.

Under the symmetric action $\ell'_i = \ell_i - \Delta_{\text{tr}}$ and $\ell'_{\text{bg}} = \ell_{\text{bg}} + \Delta_{\text{tr}}$, background beats the current owner by margin m_{bg} iff

$$\begin{aligned} \ell_{\text{bg}} + \Delta_{\text{tr}} &\geq \ell_i - \Delta_{\text{tr}} + m_{\text{bg}}, \\ \iff \Delta_{\text{tr}} &\geq \frac{\ell_i - \ell_{\text{bg}} + m_{\text{bg}}}{2}. \end{aligned} \quad (\text{A.9})$$

It beats the strongest untouched foreground competitor iff

$$\begin{aligned} \ell_{\text{bg}} + \Delta_{\text{tr}} &\geq \ell_{\text{alt}} + m_{\text{bg}}, \\ \iff \Delta_{\text{tr}} &\geq \ell_{\text{alt}} - \ell_{\text{bg}} + m_{\text{bg}}. \end{aligned} \quad (\text{A.10})$$

Together with nonnegativity, Eq. (25) is the smallest symmetric transfer satisfying both margins. Equation (26) applies its capped value and satisfies them whenever the cap does not bind.

B Teacher–Selector Details

B.1 Background and Deployment Domain

Let $\partial\Omega$ be a narrow image border. The background slot is selected from detached alpha by

$$i_{\text{bg}} = \arg \max_i \left[\frac{1}{|\partial\Omega|} \sum_{x \in \partial\Omega} \alpha_i(x) + \lambda_{\text{mass}} \frac{1}{|\Omega|} \sum_{x \in \Omega} \alpha_i(x) \right]. \quad (\text{B.1})$$

The deployment domain contains only foreground-owned pixels whose owner has sufficient hard area and peak alpha and whose required transfer does not exceed a labeling limit. These conditions are detached and are identical when producing teacher labels and applying the trained selector.

B.2 High-Precision Tri-State Evidence

For an owner i and background i_{bg} , define the individual renderer errors

$$E_i(x) = \frac{1}{3} \|\mathbf{y}_i(x) - \mathbf{x}(x)\|_2^2, \quad (\text{B.2})$$

$$E_{\text{bg}}(x) = \frac{1}{3} \|\mathbf{y}_{i_{\text{bg}}}(x) - \mathbf{x}(x)\|_2^2. \quad (\text{B.3})$$

Let $E_{\text{cur}}(x)$ and $E_{\text{act}}(x)$ be the composite errors before and after the analytic transfer. We use

$$G_{\text{act}}(x) = \frac{E_{\text{cur}}(x) - E_{\text{act}}(x)}{E_{\text{cur}}(x) + E_{\text{act}}(x) + \epsilon}, \quad (\text{B.4})$$

$$G_{\text{ind}}(x) = \frac{E_i(x) - E_{\text{bg}}(x)}{E_i(x) + E_{\text{bg}}(x) + \epsilon}, \quad (\text{B.5})$$

where positive G_{ind} favors background. Pixels for which the action is demonstrably harmful or the foreground renderer is substantially better form the protected set,

$$c_{-}(x) = \mathbb{I}[G_{\text{act}}(x) \leq \theta_{\text{act}}^{-}] \vee \mathbb{I}[G_{\text{ind}}(x) \leq \theta_{\text{ind}}^{-}]. \quad (\text{B.6})$$

The positive teacher combines two complementary high-precision branches. The first, c_{act}^{+} , requires a positive counterfactual action gain and conservative attention/alpha-density checks. The second, c_{local}^{+} , uses a local target-image background prototype to break the feedback loop in which a pixel captured by foreground never trains the background renderer. Let $i(u) = \arg \max_k \alpha_k(u)$ denote the decoded hard owner at pixel u . For a neighborhood $\mathcal{N}(x)$,

$$\boldsymbol{\mu}_{\text{local}}(x) = \frac{\sum_{u \in \mathcal{N}(x)} \mathbb{I}[i(u) = i_{\text{bg}}] \mathbf{x}(u)}{\sum_{u \in \mathcal{N}(x)} \mathbb{I}[i(u) = i_{\text{bg}}] + \epsilon}, \quad (\text{B.7})$$

$$E_{\text{local}}(x) = \frac{1}{3} \|\mathbf{x}(x) - \boldsymbol{\mu}_{\text{local}}(x)\|_2^2. \quad (\text{B.8})$$

The branch requires enough background-owned neighbors, a low normalized local error, and a feasible transfer magnitude. The raw positive confidence and conflict-resolved label are

$$\begin{aligned} c_{+}(x) &= \max\{c_{\text{act}}^{+}(x), c_{\text{local}}^{+}(x)\}, \\ \tilde{c}_{+}(x) &= c_{+}(x)[1 - c_{-}(x)]. \end{aligned} \quad (\text{B.9})$$

Positive labels do not use an object-relative vertical coordinate.

B.3 Selector Features and Ranking Loss

The raw pointwise evidence has 139 dimensions: eight ownership/logit/density scalars, a 64-dimensional owner feature, a 64-dimensional owner-background feature difference, and a three-dimensional equivariant RGB prediction difference. One shared 1×1 projection maps the owner feature to eight dimensions and, with the same weights and canceled bias, maps the owner-background difference to eight dimensions, giving a 27-dimensional selector input. With SiLU activation ϕ ,

$$\begin{aligned}\mathbf{r}_1(x) &= \phi(\mathbf{W}_1 \mathbf{f}(x) + \mathbf{b}_1), \\ \mathbf{r}_2(x) &= \phi(\mathbf{W}_2 \mathbf{r}_1(x) + \mathbf{b}_2), \\ \pi_{\text{sel}}(x) &= \sigma(\mathbf{w}_3^\top \mathbf{r}_2(x) + b_3).\end{aligned}\tag{B.10}$$

Hidden widths 32 and 16 yield 1,961 trainable parameters.

Let $\nu(x)$ be the selector logit. Positives and negatives are separately macro-averaged over each scene-owner group. Within a group, soft extrema

$$\nu_+^{\min} = -T_{\text{rank}} \log \frac{\sum_x \tilde{c}_+(x) e^{-\nu(x)/T_{\text{rank}}}}{\sum_x \tilde{c}_+(x) + \epsilon},\tag{B.11}$$

$$\nu_-^{\max} = T_{\text{rank}} \log \frac{\sum_x c_-(x) e^{\nu(x)/T_{\text{rank}}}}{\sum_x c_-(x) + \epsilon}\tag{B.12}$$

define

$$\mathcal{L}_{\text{rank}} = T_{\text{rank}} \text{softplus} \left(\frac{m_{\text{rank}} + \nu_-^{\max} - \nu_+^{\min}}{T_{\text{rank}}} \right).\tag{B.13}$$

Unknown pixels appear in neither class loss nor ranking denominator and therefore contribute zero selector gradient. With group-balanced positive and protected logistic losses $\mathcal{L}_+$ and $\mathcal{L}_-$, the complete objective is

$$\mathcal{L}_{\text{sel}} = \mathcal{L}_+ + \lambda_- \mathcal{L}_- + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}}.\tag{B.14}$$

C Implementation and Curriculum Details

C.1 Huber Loss

For a scalar residual e and threshold $\delta > 0$, the notation used in the main paper is

$$H_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| \leq \delta, \\ \delta(|e| - \frac{1}{2}\delta), & |e| > \delta. \end{cases}\tag{C.1}$$

For $\mathbf{e} \in \mathbb{R}^d$, the overlined form denotes the coordinate mean,

$$\overline{H}_\delta(\mathbf{e}) = \frac{1}{d} \sum_{q=1}^d H_\delta(e_q).\tag{C.2}$$

C.2 Discrete Scale-Steered Convolution

Let $\mathbf{H} \in \mathbb{R}^{C_{\text{in}} \times H_\ell \times W_\ell}$ be a discrete feature map and let $\mathbf{q} \in \mathbb{Z}^2$ index an output pixel. In feature-pixel units, the commanded tap dilation is

$$d_\ell(s) = \frac{s}{s_{\text{ref}}} \kappa_\ell.\tag{C.3}$$

The implemented scale-steered cross-correlation is

$$(\hat{\mathcal{C}}_s \mathbf{H})_o[\mathbf{q}] = b_o + \sum_{c=1}^{C_{\text{in}}} \sum_{\delta \in \mathcal{K}} W_{o,c,\delta} \text{Bilin}(\mathbf{H}_c, \mathbf{q} + d_\ell(s)\delta). \quad (\text{C.4})$$

Here $\mathbf{W} = [W_{o,c,\delta}]$ is the learned kernel tensor and Bilin is bilinear sampling on the current feature grid. Pixel centers are converted to normalized $[-1, 1]^2$ coordinates with aligned corner centers; samples outside the finite canvas use zero padding. When $s = s_{\text{ref}}$ and $\kappa_\ell = 1$, all taps are integer-aligned and Eq. (C.4) reduces to the corresponding ordinary stride-one convolution up to floating-point interpolation error.

Batch and slot axes are merged, but each rendered slot retains its own scalar s . The implementation constructs all $|\mathcal{K}|$ analytic tap locations directly from Eq. (C.3); no network predicts deformable offsets. A packed five-dimensional sampling layout evaluates the taps in one vectorized operation, followed by the same channel contraction as a conventional convolution. This packed implementation and the reference per-tap implementation are algebraically identical, and gradients propagate through the bilinear sampler to the input field, scale command, and bounded layer gain.

C.3 Decoder Topology

The Obj3D instantiation broadcasts each slot to 16×16 . It applies bilinear $2\times$ synthesis followed by a 5×5 scale-steered convolution at 32×32 , another bilinear synthesis followed by a 3×3 scale-steered convolution at 64×64 , and three 3×3 scale-steered blocks at the final resolution. The alpha head is 1×1 . The RGB path pointwise-fuses the equivariant field with a spatially constant appearance style before its scale-steered head; only its bounded, zero-initialized micro branch uses a fixed-pixel 3×3 convolution. Progressive synthesis avoids the checkerboard phase preference of zero-insertion while retaining learned scale-dependent aggregation after every resolution change.

The global reference scale is continued during cold start without breaking the homogeneity of Eq. (6). Abbreviate $s_c = s_{\text{cold}}$ and $s_r = s_{\text{ref}}$. With update t ,

$$s_{\text{ref}}^{\text{eff}}(t) = \begin{cases} s_c, & 0 \leq t < 2,000, \\ s_c^{1-\beta_t} s_r^{\beta_t}, & 2,000 \leq t < 10,000, \\ s_r, & t \geq 10,000, \end{cases} \quad (\text{C.5})$$

where $\beta_t = \text{clip}_{[0,1]}((t - 1999)/8000)$, s_{cold} is the RMS scale of uniform attention on the initial grid, and s_{ref} is the final operator gauge. At every fixed update t , $\rho_{\ell,t}(ks) = k\rho_{\ell,t}(s)$ still holds for every layer ℓ .

C.4 Appearance-Transplant Pairing

Each recipient scene replaces only one foreground slot, avoiding simultaneous changes to several footprints and their background competition. Pair eligibility is computed entirely from detached factual decompositions. We exclude inactive or degenerate supports, strong boundary truncation, and excessive foreground overlap. For MOVi-C, a detached factual-coherence refinement additionally requires OneFG compactness at least 1, at least 8 effective attention pixels, and IoU at least 0.6 between hard slot ownership and its gamma-2 OneFG support thresholded at 0.5. Obj3D uses no additional coherence refinement. The pair confidence ω_{ij} combines the base factual quality terms; it never observes the counterfactual outcome, so a poor transplant cannot escape supervision by invalidating its own pair.

This self-supervised factual filtering constructs training losses, where privileged input-side object cores are unavailable. The Obj3D evaluation population and reported metric denominators use the frozen input-only bank and common measurement-validity rules in Sec. D.

The compactness target in the main paper implies the coverage relation

$$A_{i \leftarrow j}^{\text{target}} = \text{sg}\left(K(m_j^{\text{don}})\right) \text{sg}(r(m_i^{\text{rec}}))^2. \quad (\text{C.6})$$

Thus recipient geometry commands spatial extent, while donor appearance may retain its fill density, transparency, and occlusion behavior.

C.5 Single-Frame Schedule

The controlled Obj3D curriculum uses one continuous 70k-step trajectory with single-frame training, random position initialization, and no optimizer or scheduler reset. Optimization uses AdamW with batch size 64, base learning rate 4×10^{-4} , zero weight decay, BF16 AMP, global gradient-norm clipping at 0.05, a 10k-step linear warm-up followed by cosine decay to zero at 70k, and decoder compilation:

- steps 0–15k: reconstruction warm-up, with the terminal-gate regularizer;
- steps 15k–40k: ramp and retain $\mathcal{L}_{\text{pos}}$ and $\mathcal{L}_{\text{ov}}$;
- steps 40k–70k: retain factual losses and add counterfactual appearance transplantation with $\mathcal{L}_{\text{geo}}$.

The locked Obj3D core coefficients are $\lambda_{\text{pos}} = 0.2$, $\lambda_{\text{ov}} = 0.01$, $\lambda_{\text{geo}} = 0.002$, $\lambda_K = 0.25$, and $\lambda_{\text{gate}} = 10^{-4}$. Auxiliary losses enter through ramps near their activation boundaries. In the main trajectory, the optional selector is jointly trained from 35k with its disjoint AdamW optimizer [Loshchilov and Hutter \(2019\)](#) and is applied from 36.2k with a ramped transfer cap. Its threshold and cap were selected on a calibration panel sampled arbitrarily from the training split; no validation or evaluation sample entered this panel. The selected operating point is then used during the same continuous training trajectory. The 40k diagnostic in Fig. 4 comes from a separate controlled run using reconstruction, $\mathcal{L}_{\text{pos}}$, and $\mathcal{L}_{\text{ov}}$ only. The diagnostic omits teacher-selector activation, counterfactual geometry, and temporal initialization. The selector’s class/ranking weights are $\lambda_- = 2$ and $\lambda_{\text{rank}} = 0.25$.

C.6 Temporal Initializer Semantics

The temporal module consumes the raw slot state $\mathbf{z} = [\mathbf{a}; \mathbf{p}; s]$ and predicts an initialization residual, not a separately evaluated future state. Its zero-initialized final projection returns $(\delta\mathbf{a}, \delta\mathbf{p}, \delta\ell)$: appearance uses an additive update, position a bounded additive update, and scale $s^{(0)} = \text{clip}(s \exp(\delta\ell))$. Thus the update is additive in log-scale even though the Transformer input is raw scale. Because a converged slot need not be the best input to a finite next-frame ISA iteration, the residual may re-noise or precondition appearance as well as adjust geometry. The time-slot Transformer [Li et al. \(2024\)](#); [Vaswani et al. \(2017\)](#) preserves the slot axis, mixes scene context through spatial attention, and returns one residual per slot. Ordered carry-forward and current-frame ISA refinement maintain identity. Next-frame reconstruction jointly trains the Transformer and final projection through current-frame ISA refinement. Freezing or jointly adapting the calibrated single-frame core is a benchmark protocol choice. The explicit update is given in main-paper Eq. (27).

C.7 Gradient Routing Summary

The following detach boundaries prevent moving-target solutions:

- $\mathcal{L}_{\text{rec}}$ updates the full core.
- $\mathcal{L}_{\text{ov}}$ keeps attention live.
- $\mathcal{L}_{\text{pos}}$ keeps the OneFG target and selected-background logit paths live but detaches the position and scale commands, background-slot identities, and factual validity mask.
- $\mathcal{L}_{\text{geo}}$ keeps the counterfactual alpha path and donor appearance live while detaching recipient commands, factual moments, pair selection, and confidence.
- teacher construction, selector inputs, and deployment eligibility are detached; $\mathcal{L}_{\text{sel}}$ updates only selector parameters through a disjoint optimizer.

D Obj3D Evaluation Protocol and Extended Results

D.1 Models, Validation Population, and Support Roles

ISA, GeoCo-SAVi, and our same-protocol SlotAug implementation [Kim et al. \(2023\)](#) train for 70k optimizer updates with training seed 42, single-frame training, and random position initialization. All results use the complete Obj3D validation set and four iid inference-initialization seeds, 20260717, 20260718, 20260719, and 20260720, totaling 10,400 scene-seed rows. These evaluation seeds are distinct from the training seed. The three methods match encoder and decoder capacity, update count, sample exposure, and the remaining comparison protocol. SlotAug retains the published loss hyperparameters and instruction semantics. Applying LayerNorm [Ba, Kiros, and Hinton \(2016\)](#) directly to this implementation’s command input collapses the command distinctions needed by the task, so we move that normalization to a later hidden layer. The reported numbers are therefore from the resulting 64×64 same-protocol implementation.

Training support is derived only from model decoder/attention outputs; neither the RGB reference bank below nor any evaluation pseudo-label enters optimization. During evaluation, the frozen RGB cores serve only as *reference support*: they select a common object-slot identity across methods before any edit. Geometry is then read from each method’s *measurement support*, the mutually exclusive full-softmax hard mask

$$\hat{m}_i(\mathbf{u}) = \mathbb{I} \left[i = \arg \max_j \alpha_j(\mathbf{u}) \right], \quad n_i = \sum_{\mathbf{u}} \hat{m}_i(\mathbf{u}). \quad (\text{D.1})$$

On the normalized grid $\mathbf{q}(\mathbf{u}) \in [-1, 1]^2$, its centroid, RMS radius, and coverage are

$$\begin{aligned} \hat{\boldsymbol{\mu}}_i &= \frac{1}{n_i} \sum_{\mathbf{u}} \hat{m}_i(\mathbf{u}) \mathbf{q}(\mathbf{u}), \\ \hat{r}_i^2 &= \frac{1}{n_i} \sum_{\mathbf{u}} \hat{m}_i(\mathbf{u}) \|\mathbf{q}(\mathbf{u}) - \hat{\boldsymbol{\mu}}_i\|_2^2, \quad \hat{A}_i = \frac{n_i}{HW}. \end{aligned} \quad (\text{D.2})$$

Obj3D supplies no instance GT masks. These quantities therefore measure how the exposed geometry controls the model’s decoded support. GT IoU, mIoU, FG-ARI, and true object motion lie outside this analysis. The external RGB core defines frozen reference identity, and decoded

support supplies the measurement mask, keeping decoder consistency separate from the reconstruction/segmentation discrepancy. Centroid errors and appearance-transplant centroid drift are L_2 distances on the $[-1, 1]^2$ normalized decoder grid. Hard radius uses RMS-radius units on the same grid, whereas coverage is the dimensionless area fraction $n_i/(HW)$. Log-slopes and monotonicity are dimensionless. Translation commands are defined on a common 64×64 pixel canvas and their errors are reported as a fraction of its diagonal; PSNR is reported in dB.

D.2 Frozen RGB Reference Bank and Common Locks

Evaluation first constructs a frozen, model-independent object bank Q_{RGB} from input RGB alone. For every video, five uniformly spaced validation frames contribute to a per-pixel background estimate $\mathbf{B}(\mathbf{u})$: pixels with chroma at most 0.035 are aggregated by a cross-video median, falling back to the ordinary median when no such sample exists. With RGB residual and chroma

$$\begin{aligned} r_t(\mathbf{u}) &= \sqrt{\frac{1}{3} \sum_{c=1}^3 (I_{t,c}(\mathbf{u}) - B_c(\mathbf{u}))^2}, \\ c_t(\mathbf{u}) &= \max_c I_{t,c}(\mathbf{u}) - \min_c I_{t,c}(\mathbf{u}), \end{aligned} \quad (\text{D.3})$$

we seed a reliable object core with $r_t(\mathbf{u}) \geq 0.035$ and $c_t(\mathbf{u}) \geq 0.045$, and define reliable background by $r_t(\mathbf{u}) \leq 0.020$. A low-chroma moving pixel may join a core candidate only within two pixels of a seed. Eight-connected components smaller than eight pixels are removed; large multi-peak contact components, one-pixel boundary bands, shadows, and low-confidence motion receive the ignore label. This gives object cores $\{C_i\}_{i=1}^{K_t}$, reliable background $B_{0,t}$, and ignore set U_t .

The input-side acceptance rule is

$$Q_{\text{RGB}}(I_t) = \mathbf{1}[1 \leq K_t \leq 5] \prod_{i=1}^{K_t} \mathbf{1}[|C_i| \geq 8] \prod_{i=1}^{K_t} \mathbf{1}[d_{\text{canvas}}(\text{bbox}(C_i)) \geq 2]. \quad (\text{D.4})$$

It is fixed before any model is evaluated and yields 2,344 object cores over the complete 2,600-row validation loader. In particular, model-dependent quantities such as alpha peak, attention cosine, and scale are never used to remove evaluation objects.

For each method m , a one-to-one Hungarian association [Kuhn \(1955\)](#) maps the frozen cores to hard supports,

$$\begin{aligned} \hat{\pi}_m &= \arg \max_{\pi \text{ one-to-one}} \sum_i |C_i \cap H_{m,\pi(i)}|, \\ \hat{j}_{m,i} &= \hat{\pi}_m(i), \end{aligned} \quad (\text{D.5})$$

where $H_{m,j}$ is the hard support of slot j . The associated core recall and observed purity are

$$\begin{aligned} \rho_{m,i} &= \frac{|C_i \cap H_{m,\hat{j}_{m,i}}|}{|C_i|}, \\ u_{m,i}^{\text{obs}} &= \frac{|C_i \cap H_{m,\hat{j}_{m,i}}|}{|H_{m,\hat{j}_{m,i}} \cap (C_i \cup B_0 \cup \bigcup_{\ell \neq i} C_\ell)|}. \end{aligned} \quad (\text{D.6})$$

The ignore set does not enter the purity denominator. Let $b_{m,i} = 1$ when method m retains an unassociated factual background slot for core i . Continuous geometry measurements use only the

common three-method layer

$$V_{\cap}(i) = \prod_{m \in \{\text{ISA}, \text{SlotAug}, \text{GeoCo}\}} \mathbf{1}[\rho_{m,i} \geq 0.5] \mathbf{1}[u_{m,i}^{\text{obs}} \geq 0.5] \mathbf{1}[b_{m,i} = 1]. \quad (\text{D.7})$$

All 2,344 cores remain in the denominator of the reference-lock eligibility rates. The single-method rates for ISA, SlotAug, and GeoCo-SAVi are 30.38%, 30.55%, and 31.06%; their common intersection is $697/2344 = 29.74\%$. The close individual rates and small intersection gap give reason to believe that no single method’s difficult cases are selectively excluded by the common layer. The absolute rate is intentionally conservative because a lock additionally requires reliable core association and an unassociated background slot; it is not an estimate of dataset-level object-discovery success. Q_{RGB} and $V_{\cap}$ define the shared reference population for the conditional Obj3D control metrics below.

D.3 Spatial Control and External Translation

The p -centroid error is measured on Q_{RGB} objects passing $V_{\cap}$ by

$$E_p = \|\mathbf{p}_i - \hat{\boldsymbol{\mu}}_i\|_2. \quad (\text{D.8})$$

Mean attention overlap is the scene average of $\sum_n w_i(\mathbf{u}_n)w_j(\mathbf{u}_n)$ over ordered $i \neq j$. For a donor–recipient pair selected from the frozen RGB bank and passing $V_{\cap}$, the transfer centroid drift is

$$E_{\text{app}} = \|\hat{\boldsymbol{\mu}}_i^{\text{edit}} - \hat{\boldsymbol{\mu}}_i^{\text{factual}}\|_2. \quad (\text{D.9})$$

For ISA and GeoCo-SAVi, this diagnostic replaces donor appearance while preserving the recipient’s explicit geometry. For SlotAug, the same standardized representation diagnostic copies the donor slot’s first appearance-dimension 64 coordinates to the recipient, while retaining the recipient’s (p_x, p_y, s) tail and every other slot. Thus all three methods share the same factual donor–recipient pairing and hard-support readout.

For a strictly shared output-space intervention, we command

$$\mathbf{d}_{\text{px}} \in \{(+6, 0), (-6, 0), (0, +6), (0, -6)\}, \quad \mathbf{d}_{\text{grid}} = \left(\frac{2d_x}{W-1}, \frac{2d_y}{H-1} \right). \quad (\text{D.10})$$

ISA and GeoCo-SAVi receive $\mathbf{p}_i \leftarrow \mathbf{p}_i + \mathbf{d}_{\text{grid}}$ without rerunning Slot Attention; SlotAug receives the corresponding native translation instruction followed by its identity instruction. With factual and edited pixel centroids $\mathbf{c}_f, \mathbf{c}_e$, the translation error is

$$E_{\text{move}} = \frac{\|(\mathbf{c}_e - \mathbf{c}_f) - \mathbf{d}_{\text{px}}\|_2}{\sqrt{H^2 + W^2}}. \quad (\text{D.11})$$

Each Q_{RGB} core anchors an ISA/SlotAug/GeoCo hard-support triple; all factual pairwise centroid distances must be at most eight pixels. To keep the shared reference identities spatially aligned, eligible triples are ranked by total factual centroid distance, and pairing cycles through the lower-distance half. Each direction then retains the common post-edit intersection for which all three methods have factual and edited supports of at least 20 pixels with a two-pixel bounding-box margin. This clipping gate is intended to reduce canvas-truncation bias. The four directions use the jointly valid command records obtained under the four inference-initialization seeds. The resulting factual, transplantation, and translation outcomes are reported once in main-paper Table 1; this supplement records their selection and measurement protocol. Translation error is averaged over the retained command records; the four seeds are repeated inference initializations.

D.4 Fixed-Scale Appearance–Size Consistency

This factor-level test writes eight matched appearances into the same target slot, fixes $\mathbf{p} = (0, 0)$ and $s = s_0$, and decodes only the target with its factual background. We form groups only from Q_{RGB} objects passing $V_{\cap}$ that admit all eight donor appearances; 249 complete groups remain. For each group and size measure x , we compute $\sqrt{\frac{1}{8} \sum_{d=1}^8 (x_d - \bar{x})^2}$, then average the group-level standard deviations equally across the 249 groups. Position, scale, and background are held fixed, so these statistics isolate residual appearance control over decoded size. ISA and GeoCo-SAVi directly fix their explicit geometry, while SlotAug uses the same fixed-geometry representation test described above. Main-paper Table 1 reports these values.

D.5 Counterfactual Scale Response

For $k \in \{0.5, 0.75, 1, 1.25, 1.5\}$, the analytic response is $\hat{r}(k)/\hat{r}(1) = k$ and $\hat{A}(k)/\hat{A}(1) = k^2$. We fit

$$\begin{aligned} \log \frac{\hat{r}(k)}{\hat{r}(1)} &= b_r + \beta_r \log k + \varepsilon_r, \\ \log \frac{\hat{A}(k)}{\hat{A}(1)} &= b_A + \beta_A \log k + \varepsilon_A, \end{aligned} \tag{D.12}$$

where b_r, b_A are fitted intercepts; the ideal slopes are one and two. We report β_r and β_A as radius log-slope and coverage log-slope. A support satisfies radius monotonicity if $\hat{r}(k_{\ell+1}) \geq \hat{r}(k_{\ell})$ for every adjacent ordered pair, and coverage monotonicity analogously. Equality from hard-mask discretization is accepted. Starting from Q_{RGB} cores, each method retains only objects whose factual and all edited supports have at least 20 pixels and a two-pixel boundary margin. This method-conditional clipping gate retains supports whose geometry is defined and not truncated by the finite canvas. It is solely a canvas-feasibility filter and does not classify whether an edit achieved its intended effect. Defined on support size and boundary margin, it does not inspect the ordered scale response; under this common predeclared rule the retained-set monotonicity is 100.0% for GeoCo-SAVi, 70.3% for SlotAug, and 23.3%/30.3% for ISA.

ISA and GeoCo-SAVi directly execute $s_i \leftarrow ks_i$ while freezing all other state. SlotAug uses its native scale instruction, followed by its identity instruction, together with the prescribed attention-centroid compensation $\mathbf{d}_i = (k - 1)\mathbf{p}_i^{\text{attn}}$. Although the control interfaces differ, every result uses the same final hard-support response definitions. The aggregate slopes and monotonicity values are reported in main-paper Table 1.

E MOVi-C Protocol and Extended Diagnostics

E.1 Common Video Protocol and Reference Implementations

All three methods are trained on MOVi-C Greff et al. (2022) for 100k optimizer updates and evaluated on the fixed official validation population of 250 videos, each containing 24 frames. RGB reconstruction and formal metrics use 64×64 frames. The frozen visual encoder is `vit_small_patch14_dinov2.lvd142m` (C et al. (2024)); its block-12 384-dimensional tokens are extracted from 336×336 inputs and projected to 128 dimensions. The reference implementations are matched in data, RGB target, encoder/decoder capacity, batch size, 100k update budget, and total sample exposure. Method-specific loss, optimizer, and curriculum hyperparameters otherwise use the published values, subject only to the adaptations disclosed below.

E.2 GeoCo-SAVi MOVi-C Architecture, Optimization, and Curriculum

Slot core and renderer. GeoCo-SAVi uses 11 slots with a 64-dimensional appearance state and explicit state $(\mathbf{a}, \mathbf{p}, s)$. It uses 64-dimensional query/key/value projections, a 64-dimensional Slot-Attention GRU [Locatello et al. \(2020\)](#); [Cho et al. \(2014\)](#), a 256-dimensional update MLP, and a 128-dimensional grid encoding. The first frame of every training clip uses three Slot-Attention iterations; a carry-forward frame uses two. Positions are initialized iid-uniformly, s is clamped to $[.001, 2]$, factor 5 normalizes the object-relative coordinate field, and the Spatially Equivariant Decoder has width 64. It forms a 32×32 canonical field, applies one scale-steered 5×5 refinement, bilinearly upsamples, then applies four scale-steered 3×3 refinements at 64×64 . The RGB head is 3×3 with sigmoid output and the alpha head is 1×1 ; there is no low-resolution MLP capacity branch. The regular steered kernel is 3×3 and its learned gain is bounded by 2.

Optimization and reference-scale continuation. We use AdamW with base learning rate 4×10^{-4} , zero weight decay, BF16 AMP, global gradient-norm clipping at 0.05, physical/effective batch size 16, and training seed 42. Only the decoder is compiled. The core rate warms up linearly for 10k updates and then follows cosine decay to zero at 100k. The training loader has eight workers, no subsampling, and uses video identity as its source key; the fixed position seed for evaluation is 20260723. Let $s_c = 0.851256531$, $s_f = 0.2$, and $\ell_t = (1 - \beta_t) \log s_c + \beta_t \log s_f$. The global numerical reference of the scale-steered operator follows

$$s_{\text{ref}}(t) = \begin{cases} s_c, & t < 2000, \\ e^{\ell_t}, & 2000 \leq t < 10000, \\ s_f, & t \geq 10000, \end{cases} \quad (\text{E.1})$$

where $\beta_t = \min((t - 1999)/8000, 1)$. This reference is a global numerical continuation for the convolution operator, not an object-size prior or a replacement for the predicted s_i .

Loss activation and temporal curriculum. Position and normalized-attention overlap activate at 15k and reach full weight over 2k updates. We report the rounded coefficients $\lambda_{\text{pos}} = 0.04$ and $\lambda_{\text{ov}} = 0.018$. Position uses coordinate Huber loss with threshold 0.05 on the detached-command gamma-2 OneFG centroid; only loose factual foreground supports with OneFG coverage at most 0.2 pass the detached validity filter, and no separate valid-background gate is applied. $\mathcal{L}_{\text{geo}}$ is kept at zero until 48k, ramps linearly to full weight by 50k, and then uses $\lambda_{\text{geo}} = 0.004$, unit center/radius terms, and a 0.25 compactness term, all with Huber threshold 0.05. Its gamma-2 OneFG pairs use factual quality gates and the receiver-center/radius, donor-compactness semantics described in Sec. 3. It is evaluated at time index 0 for two-frame clips and at indices 0 and 2 for four-frame clips.

The residual STATM initializer has width 128, four heads, a 512-dimensional MLP, and four history states. It detaches history and uses zero dropout. Its output projection is zero-initialized, so the transition at 70k starts from exact carry-forward. Its parameter group belongs to the same AdamW instance from update zero but receives no gradient before 70k; it then warms up locally through 72k and follows cosine decay to 100k with final multiplier 0.25. Selective alpha ownership is disabled in this MOVi-C result.

The STAITUS implementation uses the same-capacity SAVi encoder/decoder core, a temporal appearance predictor, and an active-set gate, with the published RGB, temporal, and separation losses. Under the matched protocol, all positive sparsity coefficients tested close most slots and produce a degenerate decomposition, so its sparsity coefficient is set to zero. It retains the published

Updates	Frames / clip	Active terms	Purpose
0–15k	2	RGB reconstruction; s_{ref} continuation	Stable slot formation with carry-forward
15–17k	2	$\mathcal{L}_{\text{pos}}, \mathcal{L}_{\text{ov}}$ ramp	Factual geometry and allocation calibration
17–48k	2	$\mathcal{L}_{\text{pos}}, \mathcal{L}_{\text{ov}}$	Core-only stabilization
48–50k	2	$\mathcal{L}_{\text{geo}}$ ramp	Counterfactual geometry activation
50–70k	2	Full core objective	Appearance-transplantation calibration
70–72k	4	Zero-residual STATM local LR warm-up	Temporal-module activation
72–100k	4	Full STATM local schedule	Joint temporal refinement

Table 5: Final GeoCo-SAVi MOVi-C curriculum. All boundaries are strict resumes of one optimizer, scheduler, and AMP-scaler trajectory.

15k all-active warm-up followed by a 15k gate ramp. SlotContrast is adapted from feature prediction to RGB prediction so that its target space matches the other methods; it uses the same-capacity encoder/decoder, RGB MSE, and the published slot–slot contrastive settings. These are same-protocol implementations for controlled comparison.

E.3 Factual Reconstruction and Instance Metrics

For each method, full-softmax alpha is bilinearly resized to the official GT resolution before taking a slotwise argmax. FG-ARI, global ARI, Hungarian mIoU, and mean best overlap (mBO) are averaged with equal weight per video; global ARI retains the GT background. RGB values lie in $[0, 1]$; PSNR is the mean per-example $10 \log_{10}(1/\text{MSE})$, with MSE averaged over that example’s RGB entries. Neither the factual metrics nor their denominators use quality filters, OneFG, explicit geometry, or edited states. Official visible instance segmentations are used only for these metrics, never as model input.

For factual attention overlap, the metric uses each method’s final Slot Attention allocation maps. GeoCo-SAVi and STAITUS denote these maps by w_i and average

$$\frac{1}{N(N-1)} \sum_{i \neq j} \sum_n w_i(\mathbf{u}_n) w_j(\mathbf{u}_n) \quad (\text{E.2})$$

over every frame of all 250 validation videos. SlotContrast uses its corresponding final grouping maps; every map is spatially ℓ_1 -normalized before the same ordered off-diagonal pair average. STAITUS is evaluated only on its factual active slots, consistent with the state that contributes to its reconstruction. This convention is favorable to STAITUS because closed slots are excluded, whereas GeoCo-SAVi’s all-slot average also includes diffuse empty-slot maps. This row is a factual allocation diagnostic, not an alpha-overlap or editable-geometry metric.

E.4 GT-Reference Lock for Geometry Interventions

Reference-lock evaluation deliberately separates factual segmentation from controlled intervention. For every official GT track, its reference frame is the frame with maximum visible-mask area, breaking ties by the earliest frame. Only tracks with at least 20 pixels are candidates. We erode

the reference GT mask by one pixel, falling back to the original mask if erosion empties it, and associate the GT object to a factual hard-alpha slot once with Hungarian matching. The locked pair must satisfy

$$\text{core recall} \geq 0.5, \quad \text{purity} \geq 0.5. \quad (\text{E.3})$$

For translation and scale, the reference GT bounding box must additionally have a two-pixel canvas margin. This construction yields 1,543 area-qualified tracks and 178 common GeoCo–STAITUS track–slot locks.

The GT mask is external *reference support*: it fixes the same identity before editing but does not supply the reported centroid, radius, or coverage. Those quantities use each locked method’s decoded hard mask as *measurement support*. Consequently, the 178-lock analysis measures conditional control given a stable shared identity; it is not a full-dataset segmentation or intervention-success rate. Factual segmentation is reported separately over all 250 videos.

The association is fixed before intervention. Edited outputs are *not* matched again to GT or to a new Hungarian assignment. The only post-edit filter is output-only: the locked hard support must have at least 20 pixels and a two-pixel bounding-box margin. Thus GT merely defines a reproducible pre-edit reference population; it cannot certify that a particular edit succeeded.

Coordinates lie on the $[-1, 1]^2$ decoder grid. Translation commands are ± 6 pixels horizontally or vertically, and their normalized error is

$$E_{\text{move}} = \frac{\|(\mathbf{c}_e - \mathbf{c}_f) - \mathbf{d}_{\text{px}}\|_2}{\sqrt{H^2 + W^2}}, \quad (\text{E.4})$$

where $\mathbf{c}_f$ and $\mathbf{c}_e$ are the factual and edited hard-support centroids. Fixed-scale tests hold the factual position and scale command constant while changing appearance, yielding fixed- s radius std. and fixed- s coverage std. Radius log-slope and coverage log-slope regress the log change in hard-support radius and coverage against the commanded log-scale; their analytic targets are one and two, and the corresponding ordered tests report radius monotonicity and coverage monotonicity. Appearance transplantation retains the recipient geometry command while substituting donor appearance, and reports transfer centroid drift. All interventions decode the frozen factual state without rerunning Slot Attention. The resulting metrics are reported once in Table 3.

F Obj3D Ablation Protocol

F.1 Scale-Steered Decoder Control

The decoder ablation uses a conventional non-equivariant renderer while retaining direct calibration of the exposed scale against factual spread. To distinguish a correct first-order radius response from density collapse, we report the ratios at a $2\times$ scale command,

$$\begin{aligned} R_r &= \frac{r(2s)}{r(s)}, & R_{\text{soft}} &= \frac{A_{\text{soft}}(2s)}{A_{\text{soft}}(s)}, \\ R_{\gamma=2} &= \frac{A_{\gamma=2}(2s)}{A_{\gamma=2}(s)}, & R_K &= \frac{K_{\text{soft}}(2s)}{K_{\text{soft}}(s)}. \end{aligned} \quad (\text{F.1})$$

Here

$$\begin{aligned} A_{\text{soft}}(s) &= \frac{1}{HW} \sum_{\mathbf{u}} \alpha_i(\mathbf{u}; s), \\ A_{\gamma=2}(s) &= \frac{1}{HW} \sum_{\mathbf{u}} m_i(\mathbf{u}; s), & K_{\text{soft}}(s) &= \frac{A_{\text{soft}}(s)}{r(s)^2 + \epsilon}, \end{aligned} \quad (\text{F.2})$$

where α_i is the decoded soft OneFG support after slot softmax, $r(s)$ is its RMS radius, and m_i is its gamma-2 proxy. An equivariant decoder has targets $(R_r, R_{\text{soft}}, R_{\gamma=2}, R_K) = (2, 4, 4, 1)$ away from canvas clipping. The conventional-decoder control obtains $(1.978, 0.708, 0.0385, 0.178)$ (main-paper Table 4). It nearly matches spread and rapidly loses both soft mass and sharp support mass, producing the observed bell-shaped scale failure.

F.2 Normalized-Attention Overlap Control

The overlap ablation removes only $\mathcal{L}_{\text{ov}}$ while retaining the remaining factual objective. Its diagnostic is the raw spatial probability dot used by the loss:

$$O_{\text{attn}} = \frac{1}{N(N-1)} \sum_{i \neq j} \sum_n w_i(\mathbf{u}_n) w_j(\mathbf{u}_n), \quad (\text{F.3})$$

where every w_i is spatially ℓ_1 -normalized. Removing $\mathcal{L}_{\text{ov}}$ yields $O_{\text{attn}} = 0.0027$, compared with 0.0007 in the full model. The increase reflects greater co-allocation of normalized attention maps; this diagnostic does not use absolute alpha mass.

F.3 Counterfactual Geometry-Loss Control

The $-\mathcal{L}_{\text{geo}}$ control preserves the scale-steered decoder, the factual position and overlap losses, the continuous optimizer trajectory, and every evaluation protocol; it sets only the counterfactual geometry-loss weight to zero. Removing it raises transfer centroid drift from 0.0158 to 0.0298 and fixed- s radius std. from 0.0089 to 0.0095. The complete two-metric comparison is reported in main-paper Table 4.

F.4 Factual Position Control

The $-\mathcal{L}_{\text{pos}}$ control retains the scale-steered decoder, normalized-overlap loss, and Phase-2 $\mathcal{L}_{\text{geo}}$. Factual position-centroid calibration is removed throughout training. At the shared 70k endpoint, its final $\mathcal{L}_{\text{geo}}$ is 0.0031, compared with 0.0018 for the full model. Hence the counterfactual objective does not by itself recover a stable geometric coordinate system: its recipient-center target requires factual position-support-centroid alignment. The main-paper table reports this common-endpoint comparison.